

FIITJEE

ALL INDIA TEST SERIES

FULL TEST – IX

JEE (Main)-2022

TEST DATE: 03-04-2022

Time Allotted: 3 Hours

Maximum Marks: 300

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts**.
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A** and **Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B** : Do any five questions out of 10 Questions.

Section-A (01 – 20, 31 – 50, 61 – 80) contains 60 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **-1 mark** for wrong answer.

Section-B (21 – 30, 51 – 60, 81 – 90) contains 30 Numerical answer type questions with answer XXXXX.XX and each question carries **+4 marks** for correct answer. There is no negative marking.

Physics

PART – A

SECTION – A (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

- A shell of mass m is at rest initially. It explodes into three fragments having masses in the ratio $2 : 2 : 1$. The fragments having equal masses fly off along mutually perpendicular directions with speed v . What will be the speed of the lighter fragment?

(A) v
 (B) $\sqrt{2} v$
 (C) $2\sqrt{2} v$
 (D) $3\sqrt{2} v$
- In a P–N Junction diode when not connected to any circuit

(A) potential is the same every where.
 (B) the P - type side is at a higher potential than the N - type side.
 (C) there is an electric field at the junction directed from the N - type side to the P-type side.
 (D) there is an electric field at the junction directed from the P - type side to the N-type side.
- Mark the correct statement

(A) The energy contained in a small volume through which an electromagnetic wave is passing oscillates with the frequency of the wave.
 (B) Magnetic energy has zero average value in a plane electromagnetic wave.
 (C) Intensity in electromagnetic wave is given by $\frac{1}{2} \epsilon_0 E_0^2 c$ where E_0 is maximum electric field.
 (D) Energy density of electromagnetic wave is given by $\frac{1}{2} \epsilon_0 E_0^2 c$.
- An electric dipole is projected with velocity $\vec{v}$ in a uniform magnetic field directed inside the plane of paper as shown. Choose the correct option. (dipole is completely inside the field)

$\vec{v}$
 $\vec{P}$

B

⊗	⊗	⊗
⊗	⊗	⊗
⊗	⊗	⊗
⊗	⊗	⊗
⊗	⊗	⊗

Case-I

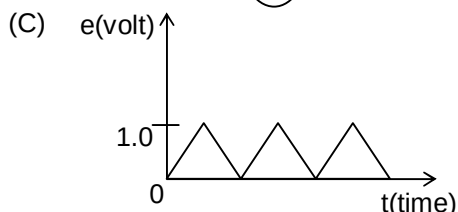
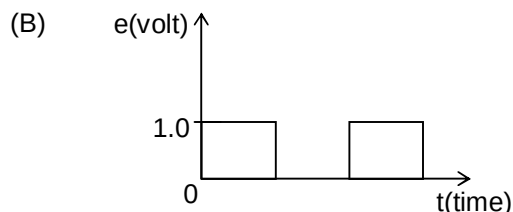
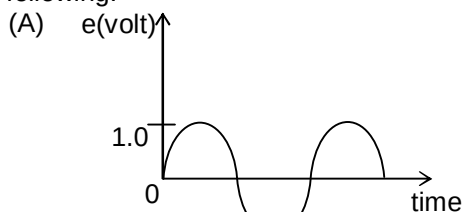
B

⊗	⊗	⊗
⊗	⊗	⊗
⊗	⊗	⊗
⊗	⊗	⊗
⊗	⊗	⊗

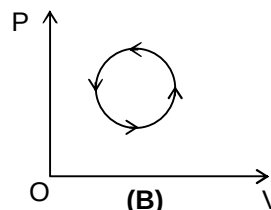
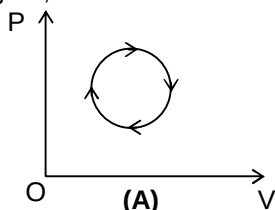
Case-II

(A) Dipole will experience a net force in case-I due to external magnetic field
 (B) Dipole will experience a net force in case-II due to external magnetic field
 (C) Dipole will experience a net torque in case-I due to external magnetic field
 (D) Dipole will experience a net torque in case-II due to external magnetic field

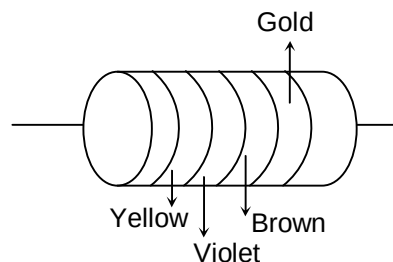
5. The time variations of signals are given as in A, B and C. Point out the true statement from the following.



- (A) A, B and C are analogue signals
 (B) A and C digital, but B is analogue signal
 (C) A and C are analogue but B is digital signal
 (D) A, B and C are digital signal
6. If the P-V diagrams of two thermodynamics devices working in a cyclic process are as shown in the figure, then:

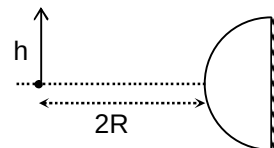


- (A) A is a heat engine, B is a heat pump/refrigerator
 (B) B is a heat engine, A is a heat pump/refrigerator
 (C) both A and B are heat engines
 (D) both A and B are heat pumps/refrigerator
7. Figure shows a colour coded carbon resistor. What is the resistance of this resistor?
- (A) $47 \times 10^1 \Omega \pm 5\%$
 (B) $23 \times 10^1 \Omega \pm 5\%$
 (C) $47 \times 10^2 \Omega \pm 5\%$
 (D) $23 \times 10^2 \Omega \pm 10\%$



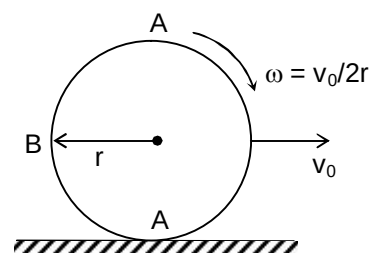
8. Select the correct statement :
- (A) Total internal reflection takes place when light travel from rarer to denser medium.
 (B) Focal length of a convex lens is positive irrespective of medium in which it is placed.
 (C) Focal length of spherical mirror depends on its radius of curvature not on the surrounding medium.
 (D) Blue colour of sky appear due to total internal reflection.

9. A glass hemisphere of radius R and material having refractive index 1.5 is silvered on its flat face as shown in the figure. A small object of height h is located at a distance $2R$ from surface of hemisphere. The final image will form :- (Assume paraxial approximation)



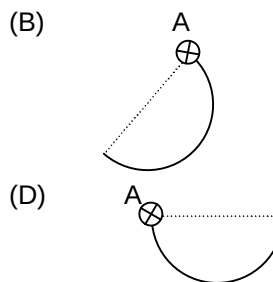
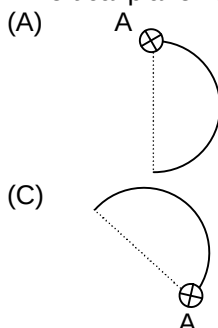
- (A) at a distance of R from silvered surface, on right side
 (B) on the hemispherical surface
 (C) at the plane surface
 (D) at a distance of $2R$ from the curved surface on left side

10. A solid sphere of mass M and radius r slips on a rough horizontal plane. At some instant it has translational velocity v_0 and rotational velocity about the centre $\frac{v_0}{2r}$. The translational velocity after the sphere starts pure rolling :



- (A) $6v_0 / 7$ in forward direction
 (B) $6v_0 / 7$ in backward direction
 (C) $7v_0 / 6$ in forward direction
 (D) $7v_0 / 6$ in backward direction

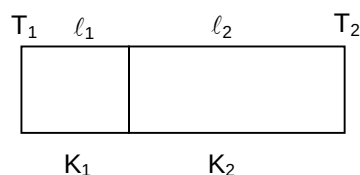
11. Which of the following figures best represents the stable equilibrium of a uniform semicircular rod in vertical plane. It is hinged at point A.



12. Two radioactive sources A and B of half lives of 2hr and 4hr respectively, initially contain the same number of radioactive atoms. At the end of 2 hours, their rates of disintegration are in the ratio

- (A) 4 : 1
 (B) 2 : 1
 (C) $\sqrt{2} : 1$
 (D) 1 : 1

13. One end of a thermally insulated rod is kept at a temperature T_1 and the other at T_2 . The rod is composed of two sections of lengths ℓ_1 and ℓ_2 and thermal conductivities K_1 and K_2 respectively. The temperature at the interface of the two sections is (cross-sectional area of both rods are same)

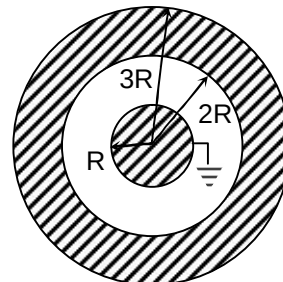


- (A) $\frac{(K_2 \ell_2 T_1 + K_1 \ell_1 T_2)}{(K_1 \ell_1 + K_2 \ell_2)}$
 (B) $\frac{(K_2 \ell_1 T_1 + K_1 \ell_2 T_2)}{(K_2 \ell_1 + K_1 \ell_2)}$

- (C) $\frac{(K_1 \ell_2 T_1 + K_2 \ell_1 T_2)}{(K_1 \ell_2 + K_2 \ell_1)}$
 (D) $\frac{(K_1 \ell_1 T_1 + K_2 \ell_2 T_2)}{(K_1 \ell_1 + K_2 \ell_2)}$

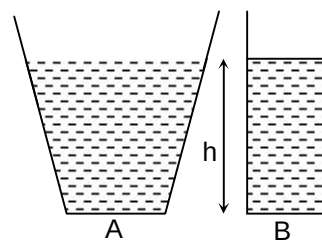
14. A conducting sphere of radius R and a concentric thick spherical shell of inner radius $2R$ and outer radius $3R$ is shown in figure. A charge $+10Q$ is given to the shell and inner sphere is earthed. Then charge on inner sphere is :

- (A) $-4Q$
 (B) $-10Q$
 (C) zero
 (D) none of these



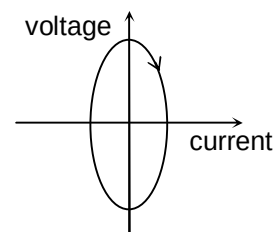
15. Two vessels A and B of different shapes have the same base area and are filled with water up to the same height h as shown in the figure. The force exerted by water on the base is F_A for vessel A and F_B for vessel B. The respective weights of the water filled in vessels are W_A and W_B . Then

- (A) $F_A > F_B$; $W_A > W_B$
 (B) $F_A = F_B$; $W_A > W_B$
 (C) $F_A = F_B$; $W_A < W_B$
 (D) $F_A > F_B$; $W_A = W_B$



16. The graph shows current v/s voltage in a series RLC a.c. circuit. The arrow indicates the direction that this curve is drawn as time progresses. In this plot, the

- (A) current lags the voltage by about 90 degrees
 (B) current leads the voltage by about 90 degrees
 (C) current and voltage are in phase
 (D) current and voltage are 180 degrees out of phase



17. When monochromatic light falls normally on the surface of the soap film in air, the intensity of the reflected light depends on the wavelength: it has a maximum at $\lambda_1 = 630\text{nm}$ and the minimum at $\lambda_2 = 525\text{ nm}$. What can be the film thickness? Refractive index of the film $n = 4/3$.

- (A) 590.625 nm
 (B) 640 nm
 (C) 562.625 nm
 (D) 466.66 nm

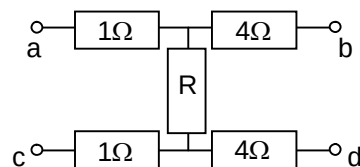
18. In a resonance tube experiment, an 80 cm air column is in resonance with a tuning fork in first overtone. Which equation can represent correct pressure variation in the air column ($x = 0$ is the top point of the tube, neglect end correction, speed of sound = 320 m/sec)

- (A) $P_0 \sin\left(\frac{15\pi x}{8}\right) \cos(600\pi t)$
 (B) $P_0 \cos\left(\frac{15\pi x}{8}\right) \sin(600\pi t)$
 (C) $P_0 \cos\left(\frac{15\pi x}{8}\right) \sin(300\pi t)$

(D) $P_0 \sin\left(\frac{15\pi x}{8}\right) \sin(600\pi t)$

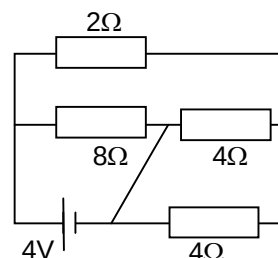
19. In the circuit, when the voltage between a and c is 20 V, the voltage between b and d is 10V. When the voltage between b and d is 20 V, what is the voltage between a and c?

- (A) 4V
(B) 5 V
(C) 8V
(D) 10V



20. In the circuit, the electric (in ampere) current through the battery is

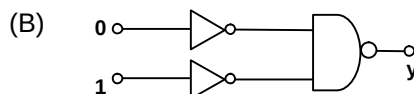
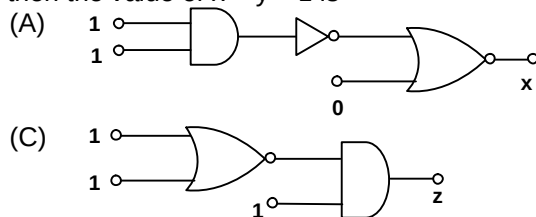
- (A) 1
(B) 2.5
(C) 4
(D) 1.5



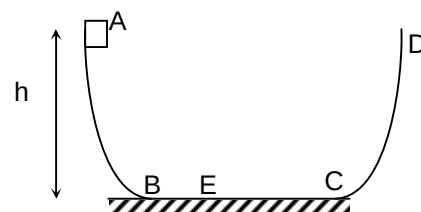
SECTION – B (Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

21. The acceleration due to gravity (g) at a place is determined by using a simple pendulum of length $\ell = (100 \pm 0.1)$ cm. If the time period is $T = (2 \pm 0.01)$ s. Then maximum percentage error in the measurement of g .
22. 1000 small mercury (Hg) drops each of the radius r and charge q coalesce together to form one spherical drop. The potential of the bigger drop is 10^n times larger than of the smaller ones then the value of n is
23. In the following combination of logic gates, the outputs of A, B and C are x , y and z respectively then the value of $x + y + z$ is



24. A small block slides along a track with elevated ends and a flat central part as shown in the figure. The flat portion BC has a length $\ell = 3.0$ m. The curved portions of the track are frictionless. For the flat part, the coefficient of kinetic friction is $\mu_k = 0.20$, the particle is released at point A which is at height $h = 1.5$ m above the flat part of the track. The block finally comes to rest. Find its distance BE where it stops.



25. In the atmosphere of a planet there are only two gases Helium and Oxygen in the mass ratio 1 : 4 respectively. The value of $\gamma = \frac{C_P}{C_V}$ of the mixture is given by $\gamma = 1 + \frac{10K}{11}$. Find the value of K.
26. If the ratio of rms speed of molecules ($\gamma = 3/2$) and speed of sound in the gas at the same temperature is x, then find the value of x. ($\sqrt{2} = 1.41$)
27. A direct current of 2A and an AC of peak value 2A flows through resistances 2Ω and 1Ω respectively. Find the ratio of heat $\left(\frac{H_{ac}}{H_{dc}}\right)$ produced in the two resistances in same interval.
28. The current in a coil of self inductance 2.0 H is increasing according to $I = 2 \sin(t^2)$ ampere. Find the amount of energy (in joule) spent during the period when the current changes from zero to 2A.
29. The difference in angular momentum associated with electron in two successive orbit of hydrogen atom is $\frac{xh}{\pi}$. Find the value of x.
30. A small fish, 0.4 m below the surface of lake, is viewed through a simple converging lens of focal length 3m. The lens is kept at 0.2 m above the water surface such that the fish lies on the optical axis of the lens. Find the image from optical centre of lens of the fish seen by the observer. The refractive index of water is $\left(\frac{4}{3}\right)$.

Chemistry

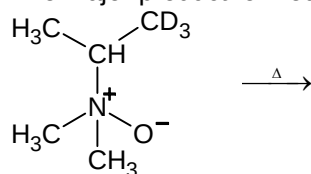
PART – B

SECTION – A (One Options Correct Type)

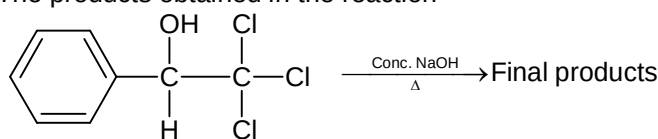
This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. Which statement is wrong?
 (A) White lung cancer is caused by textiles.
 (B) Depletion of ozone layer causes skin cancer.
 (C) PAN is a secondary pollutant.
 (D) London smog is nitrogenous smog.
32. Which of the following type of polymers have the weakest force of interactions?
 (A) Thermosetting plastic
 (B) Thermoplastic
 (C) Elastomers
 (D) Fibers
33. Which of the following compounds does not go mutarotation?
 (A) Glucose
 (B) Sucrose
 (C) Ribose
 (D) Fructose

34. The major product formed in the following reaction is:

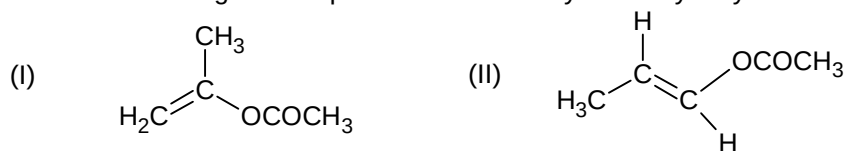


- (A) $\begin{array}{c} \text{H}_3\text{C}-\text{C}-\text{CD}_3 \\ || \\ \text{N}^+ \\ | \\ \text{H}_3\text{C}-\text{N}-\text{CH}_3 \end{array}$
- (B) $\text{H}_3\text{C}-\text{CH}=\text{CD}_2$
- (C) $\text{CH}_3-\text{N}=\text{CH}_2$
- (D) $\text{H}_2\text{C}=\text{CH}-\text{CD}_3$
35. The products obtained in the reaction



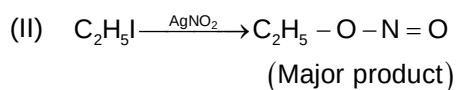
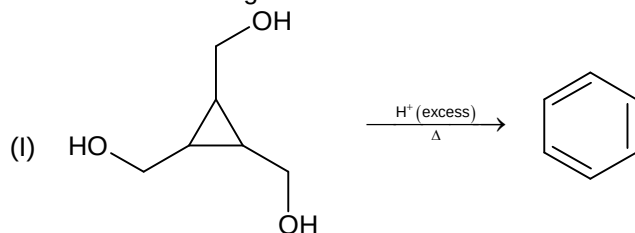
- (A) $\text{C}_6\text{H}_5\text{CHO}$ and CH_2Cl_2
- (B) $\text{C}_6\text{H}_5\text{CHO}$, HCOO^-Na^+
- (C) $\text{PhCH}_2\text{O}^-\text{Na}^+$, HCOO^-Na^+
- (D) PhCOONa , PhCH_2OH and HCOO^-Na^+

36. How can we distinguish the products obtained by acidic hydrolysis of

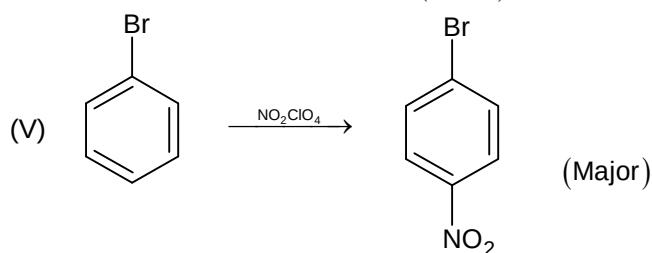
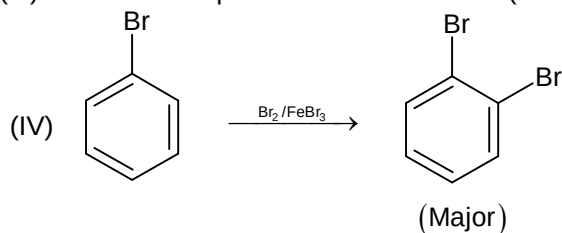


- (A) Fehling's solution
(B) NaHSO_3
(C) Brady's reagent
(D) Lucas test

37. Which of the following is incorrect?

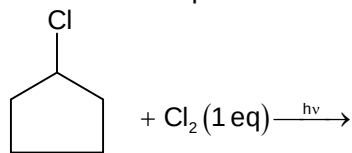


- (III) Benzene > Naphthalene > Anthracene (decreasing order of aromaticity)



- (A) II, IV
(B) I, III, V
(C) I, II, IV
(D) IV, V

38. Number of compounds formed (including stereoisomers)

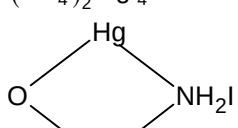


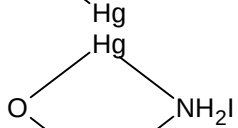
- (A) 4
(B) 5
(C) 6
(D) 7

39. Among KO_2 , AlO_2^- , BaO_2 and NO_2^+ unpaired electron present in
- KO_2
 - NO_2^+ and BaO_2
 - KO_2 and AlO_2^-
 - BaO_2
40. Which is correct among the following?
- $\text{Li} < \text{Na} < \text{K} < \text{Rb} < \text{Cs}$ (according to density)
 - $\text{B} < \text{Al} < \text{Ga} < \text{In} < \text{Tl}$ (according to atomic radii)
 - $\text{BF}_3 < \text{BCl}_3 < \text{BBr}_3 < \text{BI}_3$ (acidic nature)
 - $\text{LiF} > \text{NaF} > \text{KF} > \text{RbF} > \text{CsF}$ (solubility)
41. Which can be used to distinguish H_2O_2 and O_3 ?
- PbS
 - Starch and iodine
 - KMnO_4
 - Bleaching powder
42. Which of the following are correct?
- In $[\text{CoF}_6]^{-3}$ complex; 0 unpaired electron
 - $[\text{NiCl}_4]^{-2}$ and $[\text{Ni}(\text{CO})_4]$ both are tetrahedral
 - $[\text{Ni}(\text{NH}_3)_6]^{+2}$ is a outer complex
 - 3 moles of ions are produced by 1 mole of $\text{K}_2[\text{PtCl}_6]$
- II, III, IV
 - I, II, IV
 - I, III, IV
 - I, IV
43. Chemical volcano is produced on heating?
- NH_4NO_3
 - $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$
 - NH_4NO_2
 - $\text{K}_2\text{Cr}_2\text{O}_7$
44. $\text{HgCl}_2 + \text{KI}(\text{excess}) \longrightarrow \text{X} \xrightarrow[\text{NaOH}]{\text{NH}_4\text{Cl} +} \text{Y}$; X and Y are:
- X
 HgI_2

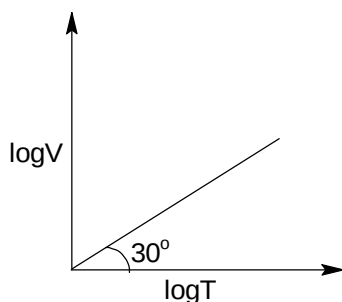
Y
 NH_4I
 - X
 K_2HgI_4

Y
 $(\text{NH}_4)_2\text{HgI}_4$
 - X
 K_2HgI_4

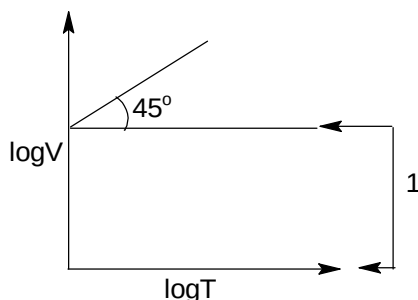
Y

 - X
 HgI_2

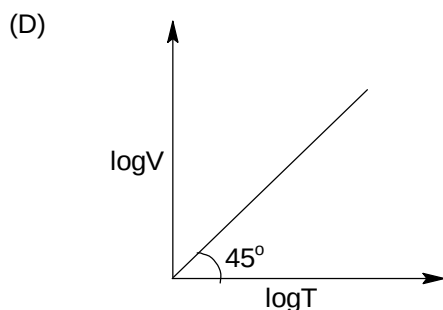
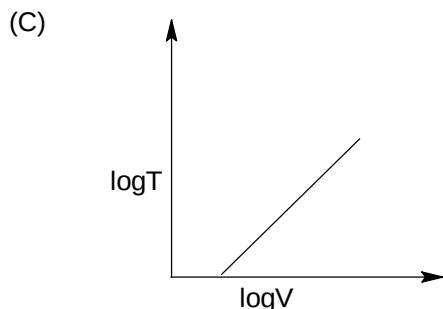
Y


45. Two closed vessel A and B of equal volume containing air at pressure P_1 and temperature T_1 are connected to each other through a narrow open tube. If the temperature of one is now maintained at T_1 and other at T_2 (where $T_1 > T_2$) then what will be the final pressure?
- (A) $\frac{T_1}{2P_1T_2}$
- (B) $\frac{2P_1T_2}{T_1 + T_2}$
- (C) $\frac{2P_1T_2}{T_1 - T_2}$
- (D) $\frac{2P_1}{T_1 + T_2}$
46. Which of the following has been arranged in order of decreasing freezing point?
- (A) $0.05\text{M NaNO}_3 > 0.04\text{M CaCl}_2 > 0.140\text{M glucose} > 0.075\text{M ZnSO}_4$
- (B) $0.04\text{M CaCl}_2 > 0.140\text{M glucose} > 0.075\text{M ZnSO}_4 > 0.05\text{M NaNO}_3$
- (C) $0.075\text{M ZnSO}_4 > 0.140\text{M glucose} > 0.04\text{M CaCl}_2 > 0.05\text{M NaNO}_3$
- (D) $0.075\text{M ZnSO}_4 > 0.05\text{M NaNO}_3 > 0.14\text{M glucose} > 0.04\text{M CaCl}_2$
47. Colloidal solutions of gold prepared by different methods are of different colours because of
- (A) different diameters of colloidal gold particles
- (B) variable valency of gold
- (C) different concentration of gold particles
- (D) impurities produced by different methods
48. Which is correct from the followings?
- (A) Packing efficiency of FCC is 74%
- (B) F-centre crystal is coloured due to excess electron is trapped in tetrahedral voids
- (C) When Ge is doped with As, then electrical conductivity decreases.
- (D) Bragg's equation is $n\lambda = d\cos\theta$
49. For a closed container $n = 200$ mole of an ideal gas fitted with movable, frictionless, weightless piston operating such that pressure of gas remains constant at 16.42 atm. Which graph represents correct variation of $\log V$ vs $\log T$
- (A)



(B)





50. Which salt from the following is treated with NaOH, produced insoluble hydroxides but soluble in aqueous NaOH?
- (A) FeCl_3
 (B) HgCl_2
 (C) ZnSO_4
 (D) MnSO_4

SECTION – B (Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXX.XX).

51. 3 g sample of peanut oil is mixed to 25 ml of 0.40 M NaOH. After complete saponification, 4 ml of 0.28 M H_2SO_4 is added to neutralize excess of NaOH. The saponification number of peanut oil is [Given : Saponification number is milligrams of NaOH or KOH used by 1 g oil]
52. If radiation corresponds to second line of “Balmer series” of Li^{+2} ion knocked out electron from first excited state of H-atom then kinetic energy of ejected electron would be in eV?
53. The nucleidic ratio of ${}^3_1\text{H}$ to ${}^1_1\text{H}$ in a sample of water is $6.0 \times 10^{-18} : 1$. Tritium undergoes decay with half life period of 12.3 years. If $x \times 10^6$ tritium atoms are present from 10 g of such sample after 40 years after the original sample collected. [Given: $\log 9.506 = 0.978$, Avagadro’s number = 6.022×10^{23}].
54. What is the pH at which 1.25×10^{-4} M Co^{2+} ions in solution precipitate as $\text{Co}(\text{OH})_2$? (K_{sp} of $\text{Co}(\text{OH})_2$ is 2.5×10^{-16}) [Given : $\log 1.414 = 0.1504$].

55. Calculate the heat produced (in kJ) when 112 g of CaO is completely converted to CaCO_3 by the reaction with CO_2 at 27°C in a container of fixed volume.
Given: $\Delta H_f^\circ [\text{CaCO}_3, (\text{s})] = -1207 \text{ kJ/mol}$
 $\Delta H_f^\circ [\text{CaO}, (\text{s})] = -635 \text{ kJ/mol}$
 $\Delta H_f^\circ [\text{CO}_2 (\text{g})] = -394 \text{ kJ/mol}$
 $R = 8.3 \text{ JK}^{-1} \text{ mol}^{-1}$
56. 50 ml of 0.1 M $\text{CuSO}_4 (\text{aq})$ solution was electrolysed using inert electrodes by passing current till the pH of the resulting solution was 2. The solution after electrolysis was neutralized and then reacted with excess KI and formed iodine (I_2) titrated with 0.04 M $\text{Na}_2\text{S}_2\text{O}_3$. Calculate the required volume (in ml) of $\text{Na}_2\text{S}_2\text{O}_3$.
57. 0.535 g ethanol and acetaldehyde mixture when heated with Fehling's solution gave 1.2 g of a red precipitate. What is the percentage of ethanol in the mixture? [At. Wt. of Cu = 63.5 g].
58. In an evacuated vessel of capacity 110 litres, 4 moles of Argon and 5 moles of PCl_5 were introduced and equilibrated at a temperature 250°C . At equilibrium, the total pressure of the mixture was found to be 4.678 atm. Calculate K_p for the reaction
 $\text{PCl}_5 \rightleftharpoons \text{PCl}_3 + \text{Cl}_2$ at this temperature.
59. What is the percentage vacant space in one layer of square packing of spheres touching each other having 4 spheres?
60. The optical rotation of the α – form of a pyranose is $+150.7^\circ$ and β – form is $+52.8^\circ$. In solution an equilibrium mixture of these anomers has an optical rotation of $+75.7^\circ$. The percentage of the α – form in equilibrium mixture is

Mathematics

PART – C

SECTION – A (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. $\int \left(\frac{e^{\sin x}}{e^{\sin x} + x} \right)^2 \left(\frac{1}{x} - \cos x \right) dx = \log_e (f(x)) + g(x) + c$; where c is constant, then $f(\pi) - g(\pi)$ is
- (A) $\frac{\pi+1}{\pi-1}$
 (B) $\frac{\pi-1}{\pi+1}$
 (C) $\frac{\pi+1}{\pi}$
 (D) none of these
62. In $\triangle ABC$, a semicircle is inscribed touching sides AC and BC internally and having its extremities on side AB. If radius of semicircle is 4 and length of angle bisector through $\angle C$ is 8, then $\angle C$ is
- (A) $\frac{\pi}{6}$
 (B) $\frac{\pi}{4}$
 (C) $\frac{\pi}{3}$
 (D) $\frac{5\pi}{12}$
63. If the number of matrices $A = [a_{ij}]_{2 \times 2}$ where $a_{ij} \in \{-2, -1, 0, 1, 2\}$ such that $\sum a_{ij} = 0$ is k , then
- (A) $k = 85$ if A is non skew symmetric matrix
 (B) $k = 80$ if A is non skew symmetric matrix
 (C) $k = 75$ if A is non skew symmetric matrix
 (D) $k = 65$ if A is non skew symmetric matrix
64. The number of points in $[0, 2]$ where the function $f(x) = \begin{cases} \left\lceil \tan \frac{\pi}{4} x \right\rceil & ; x < 1 \\ |3x - 4| [x] & ; x \geq 1 \end{cases}$ is either discontinuous or non differentiable is/are (where $[.]$ denotes greatest integer function)
- (A) 3
 (B) 4
 (C) 5
 (D) 2
65. If the numbers 1, 2, 3, 4, 5, 6 are arranged in a circle, then the probability that the sum of any two consecutive numbers must not be equal to 7 is
- (A) $\frac{2}{15}$

- (B) $\frac{1}{5}$
 (C) $\frac{4}{15}$
 (D) $\frac{1}{3}$

66. If $(\cos x \cot y - \sin x \tan y)dy + (\cos x \ln \cos y - \sin x \ln \sin y) dx$; $0 < y < \frac{\pi}{2}$. If $y\left(\frac{\pi}{2}\right) = \frac{\pi}{4}$, then the value of $y(0)$ is
 (A) $\frac{\pi}{8}$
 (B) $\frac{\pi}{6}$
 (C) $\frac{\pi}{4}$
 (D) $\frac{3\pi}{8}$
67. The area bounded by the curve $x = \log_e y$, $x = e^y - 2$, $x = e - e^y$, $x = -1$ is $A + B \ln\left(1 + \frac{e}{2}\right)$, then $|A + B|$ is
 (A) $2 + \frac{1}{e}$
 (B) $2 + e - \frac{1}{e}$
 (C) 2
 (D) $\frac{1}{e}$
68. If $\sum_{r=1}^n \frac{r^2 + 2r - 1}{(r+1)!} = 3 - \frac{103}{101!}$, then the value of n is
 (A) 99
 (B) 100
 (C) 101
 (D) 103
69. If the planes $x - y - z = 0$, $x - y + \sin \alpha \cdot z = 0$ and $x + \sin \alpha \cdot y - z = 0$ pass through a straight line, then number of values of α in $[0, 2021\pi]$ is/are
 (A) 1010
 (B) 1011
 (C) 2020
 (D) 2021

70. Let T be the set of isosceles triangles in which a circle of radius r can be inscribed. If the least perimeter of all such triangles possible in T is 18, then the value of r^2 is
- (A) $\frac{3}{2}$
 (B) 2
 (C) 3
 (D) 6
71. The mean and standard deviation of the marks of 20 students were found to be 40 and 15 respectively. Later, it was discovered that a score of 60 was wrongly read as 40, then the correct variance will be
- (A) 232
 (B) 236
 (C) 240
 (D) 244
72. Which of the following is a tautology?
- (A) $p \wedge (p \vee q)$
 (B) $p \vee (p \wedge q)$
 (C) $(p \wedge (p \rightarrow q)) \rightarrow q$
 (D) $q \rightarrow (p \wedge (p \rightarrow q))$
73. If the pair of straight line $x^2 - 2cxy - 7cy^2 = 0$; c is parameter intersect the parabola $y^2 = 4x$ also at points A and B other than origin, then the locus of the point of intersection of tangents drawn at points A and B on the parabola $y^2 = 4x$ is
- (A) $7y + x = 0$
 (B) $7x + 2y = 0$
 (C) $7x + 2 = 0$
 (D) $2x + 7 = 0$
74. Let $y = 2x$ be one of the diagonal of a parallelogram inscribed in an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$; $a > b$ having eccentricity $\frac{\sqrt{a^2 - 8}}{a}$. If the maximum area of parallelogram is $16\sqrt{2}$, then the value of a is
- (A) $2\sqrt{2}$
 (B) 4
 (C) $4\sqrt{2}$
 (D) 8
75. If there does not exist any chord drawn from the point $\left(2\sqrt{2} \sin \theta, \frac{\cos \theta}{\sqrt{3}}\right)$ to the circle $\sqrt{3}(x^2 + y^2) = 2\sqrt{6} \sin \theta + \cos \theta y$ which are bisected by x -axis, then
- (A) $\theta \in \left(\frac{n\pi}{3}, n\pi\right); n \in \mathbb{Z}$
 (B) $\theta \in \left(n\pi - \frac{\pi}{3}, n\pi + \frac{\pi}{3}\right); n \in \mathbb{Z}$
 (C) $\theta \in \left(n\pi - \frac{\pi}{6}, n\pi + \frac{\pi}{6}\right); n \in \mathbb{Z}$
 (D) $\left(n\pi + \frac{\pi}{6}, n\pi + \frac{\pi}{2}\right); n \in \mathbb{Z}$

76. Let the tangent drawn at any point $P(t)$ on the curve $y^2 = 4ax$; $a > 0$ having focus at S intersects the axis at point T . If the locus of the orthocentre of ΔPST intersects $y^2 = 4ax$ at points A and B other than origin, then $(AB)^2$ is
- (A) $3a^2$
(B) $12a^2$
(C) $36a^2$
(D) $48a^2$
77. The vector $\vec{a} = \hat{i} + 2\hat{j} + \hat{k}$ lying in the plane of the vectors $\vec{b} = \hat{i} + \hat{j}$ and $\vec{c}$ bisects the angle between $\vec{b}$ and $\vec{c}$. If $|\vec{c}| = \sqrt{2}$, then $\vec{c} \cdot \vec{a}$ is
- (A) 1
(B) 2
(C) 3
(D) $\frac{\sqrt{3}}{2}$
78. A tower subtends angles θ , 2θ , 3θ at points A , B and C respectively all lying on a horizontal line through the foot of tower. If $\frac{AB}{BC} = 1 + \sqrt{2}$, then θ is equal to
- (A) $\frac{\pi}{8}$
(B) $\frac{\pi}{10}$
(C) $\frac{\pi}{12}$
(D) $\frac{\pi}{15}$
79. If R be a relation defined as $R = \{(x, y) : x + y = p \text{ for some rational } p; x, y \text{ are real numbers}\}$, then R is
- (A) reflexive but not symmetric
(B) reflexive but not transitive
(C) only reflexive
(D) equivalence
80. Let $f(x) = -4 \cdot e^{\frac{2-x}{4}} + 10 + 2x + x^3 + x^5$ and $g(x) = f^{-1}(x)$, then the value of $g'(50)$ is
- (A) $\frac{1}{92}$
(B) $\frac{1}{2}$
(C) $\frac{1}{95}$
(D) $\frac{1}{94}$

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXX.XX).

81. One root of the equation $x^4 + 2x^3 + ax^2 + bx + c = 0$ is $\tan 15^\circ$. If all the roots are real and a, b, c are rational parameter, then minimum value of $b - a$ is
82. The polynomial $f(z) = az^{4036} + bz^{4035} - cz^{4034}$ has real coefficients where a, b, c belongs to $(-\infty, 4038]$. If $f\left(e^{-\frac{i\pi}{3}}\right) = -4032 + 4038\sqrt{3}i$, then the value of $|a - b|$ is
83. If $\sum_{r=0}^{\infty} \left(\frac{1}{4r+1} - \frac{1}{4r+3} \right) = L$, then $[4L]$ is (where $[.]$ denotes greatest integer function)
84. Let $a = \sin^{-1} \sin 26$, $b = \cos^{-1} \cos 36$, $c = \cos^{-1} \cos(a + b)$, then the largest integer x for which $|x + 10 - c| \leq \cos^{-1} \cos x$ is
85. Let $f(x)$ be a polynomial satisfying $\lim_{x \rightarrow -\infty} \frac{x^5 f(x)}{x^{10} + 10} = 10$, $f(2) = 3$, $f(3) = 7$, $f(4) = 13$, $f(-5) = 31$, $f(-1) = 3$, then $\lim_{x \rightarrow -5} \frac{f(x) + x - x^2 - 1}{(x+5)(x+4)(x+3)}$ is
86. Let $f(x) = \frac{(2022)^{x-1}}{(2022)^{x-1} + 1}$, then the value of $\sum_{r=1}^{4043} f\left(\frac{r}{2022}\right)$ is
87. The number of solution(s) of the equation $\sin\left(3x - \frac{\pi}{6}\right) + 2\cos^2\left(x + \frac{5\pi}{18}\right) + 1 = 0$ in $[0, 2022\pi]$
88. If $\sum_{r=1}^n [\log_2 r] = 1898$ (where $[.]$ denotes greatest integer function), then number of zeroes at the end of $n!$ is/are
89. Let N be the total number of six digits number formed which consists the digit $\{0, 1, 2, 3\}$ at least once and whose sum of digits is 10, then the sum of even divisors of N is
90. If $\sum_{k=0}^{10} \frac{{}^{10}C_k {}^{10+k}C_k}{(-2)^k} = \frac{p}{q}$ where p and q are coprime, then the value of $p + q$ is